affected by whether the concurrency is achieved through task switching or by genuine hardware concurrency. But as you may imagine, how you make use of concurrency in your application may well depend on the amount of hardware concurrency available. This is covered in chapter 8, where I cover the issues involved with designing concurrent code in C++.

1.1.2. Approaches to concurrency

Imagine for a moment a pair of programmers working together on a software project. If your developers are in separate offices, they can go about their work peacefully, without being disturbed by each other, and they each have their own set of reference manuals. However, communication is not straightforward; rather than just turning around and talking to each other, they have to use the phone or email or get up and walk to each other's office. Also, you have the overhead of two offices to manage and multiple copies of reference manuals to purchase.

Now imagine that you move your developers into the same office. They can now talk to each other freely to discuss the design of the application, and they can easily draw diagrams on paper or on a whiteboard to help with design ideas or explanations. You now have only one office to manage, and one set of resources will often suffice. On the negative side, they might find it harder to concentrate, and there may be issues with sharing resources ("Where's the reference manual gone now?").

These two ways of organizing your developers illustrate the two basic approaches to concurrency. Each developer represents a thread, and each office represents a process. The first approach is to have multiple single-threaded processes, which is similar to having each developer in their own office, and the second approach is to have multiple threads in a single process, which is like having two developers in the same office. You can combine these in an arbitrary fashion and have multiple processes, some of which are multithreaded and some of which are single-threaded, but the principles are the same. Let's now have a brief look at these two approaches to concurrency in an application.

Concurrency with Multiple Processes